

3(a)(i)	no loss of <u>kinetic</u> energy	B1
3(a)(ii)	• <u>molecules</u> have negligible volume (compared with gas/container) • no forces between <u>molecules</u> (except during collisions) • <u>molecules</u> are in random motion • collisions are instantaneous <i>Any two points, 1 mark each</i>	B2
3(b)(i)	$2mu$	A1
3(b)(ii)	$2L / u$	A1
3(b)(iii)	force = change in momentum / time = $2mu / (2L / u)$ $= mu^2 / L$	A1
3(b)(iv)	pressure = force / area = $(mu^2 / L) / L^2$ $= mu^2 / L^3$	A1
3(c)	$pV = NkT$	C1
	$NkT = \frac{1}{3}Nm\langle c^2 \rangle$ leading to $\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$ <u>and</u> $\frac{1}{2}m\langle c^2 \rangle = E_K$	A1
3(d)	$\frac{1}{2} \times 3.34 \times 10^{-27} \times \langle c^2 \rangle = (3/2) \times 1.38 \times 10^{-23} \times (25 + 273)$	C1
	r.m.s. speed = $1.9 \times 10^3 \text{ m s}^{-1}$	A1